



[FIT1047_S2_2025](#) / [Assessments](#) / [1. Written](#) / Assignment 1 Part 3 In-class test

Status	Finished
Started	Tuesday, 19 August 2025, 6:17 PM
Completed	Tuesday, 19 August 2025, 6:43 PM
Duration	26 mins 21 secs
Marks	22.50/22.50
Grade	7.50 out of 7.50 (100%)

Information

This test must be completed **in person, in your allocated class**.

We will check attendance. **Any tests attempted outside of the classroom will be reported as a breach of academic integrity.**

Open Notes Instructions

You are allowed to bring and read any notes during the test (can be paper-based or digital notes, e.g. PDF, OneNote etc). An ASCII table will be provided to you. However, you are NOT allowed to use any calculator, any other software (e.g. Logisim, Excel, language translator), any service platform (e.g. Google, ChatGPT), any communication tool (e.g. email, SMS, Whatsapp, WeChat) during the quiz. You may also bring a few pieces of blank papers for you to write down the draft or calculation steps, though we will not collect or mark these papers. All answers must be entered in Moodle.

Question 1

Complete

Not graded

Type your attendance CODE

Answer:

The correct answer is: 123



Question 2

Correct

Mark 7.50 out of 7.50

Character encodings

Complete the questions below based on the following encoding table.

The table contains 128 different characters, numbered from 0 to 127. Note how the characters of the English alphabet are not listed in the usual order. However, the order of lower-case and upper-case characters is the same: The table lists the upper-case characters first (starting with H, M, G, ...), followed by some symbols and then the lower-case characters (h, m, g, ...).

For each character, you are given its hexadecimal code. For example, the character] has hex code 2a.

!	00	T	20	e	40	...	60
@	01	B	21	x	41	æ	61
#	02	J	22	n	42	“	62
\$	03	K	23	v	43	‘	63
%	04	L	24	i	44	«	64
^	05	D	25	p	45	»	65
&	06	F	26	a	46	™	66
*	07	{	27	z	47	£	67
(	08	}	28	y	48	¢	68
)	09	[	29	r	49	€	69
-	0a	]	2a	s	4a	∞	6a
+	0b	ƒ	2b	w	4b	§	6b
=	0c	\	2c	u	4c	¶	6c
H	0d	:	2d	t	4d	•	6d
M	0e	;	2e	b	4e	≡	6e
G	0f	0	2f	j	4f	◊	6f
C	10	1	30	k	50	≠	70
Q	11	2	31	l	51	±	71
O	12	3	32	d	52	‡	72
E	13	4	33	f	53		73
X	14	5	34	"	54		74
N	15	6	35	<	55	˘	75
V	16	7	36	>	56	˙	76
I	17	8	37	,	57	¿	77
P	18	9	38	.	58	Ú	78
A	19	0	39	/	59	Æ	79
Z	1a	h	3a	?	5a	µ	7a
Y	1b	m	3b	`	5b	©	7b
R	1c	g	3c	~	5c	®	7c
S	1d	c	3d	≤	5d	¬	7d
W	1e	q	3e	≥	5e	√	7e
U	1f	o	3f	÷	5f	≈	7f

Encoding and number systems

Convert the letters of the word "Lynx" into the corresponding character codes from the table above. The final letter has already been converted for you. You can look up the hexadecimal code in the table. Then convert the code into the different number representations (decimal, binary) by hand.

Enter each number **without leading zeros**, and **without type indicators** such as 0x or 0b.

Character	Hex	Decimal	Binary
L	24 ✓	36 ✓	100100 ✓
y	48 ✓	72 ✓	1001000 ✓
n	42 ✓	66 ✓	1000010 ✓
x	41	65	1000001



Decoding

Decode the following 4-letter word. Each character is given in binary notation. Convert the binary to hexadecimal and then look up the character in the encoding table above.

Binary	Hex	Character
100001	21	B
1000000	40	e
1000110	46	a
1001001	49	r

Question 3

Correct

Mark 7.50 out of 7.50

Two's Complement

Given are two unsigned 7-bit binary numbers:

 $x = 0101110$
 $y = 1011101$

Calculate $x - y$ using 8-bit two's complement binary numbers. The first row is pre-filled with the binary representation of x . In the second row, enter the 8-bit two's complement representation of $-y$. In the third row, enter the carry bits for the addition. The final row is for the result.

x		0	0	1	0	1	1	1	0
-y		1	0	1	0	0	0	1	1
		✓	✓	✓	✓	✓	✓	✓	✓
+carry	0	0	1	0	1	1	1	0	
	✓	✓	✓	✓	✓	✓	✓	✓	
=		1	1	0	1	0	0	0	1
		✓	✓	✓	✓	✓	✓	✓	✓

Enter the result written as a decimal number:

-47



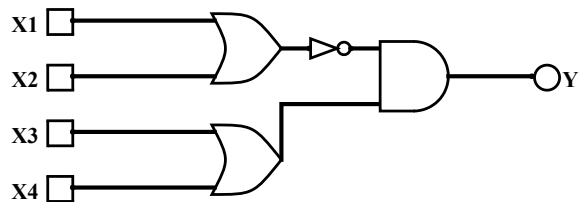
Question 4

Correct

Mark 7.50 out of 7.50

Simple Circuits 1

Given is this circuit with inputs X1, X2, X3, X4 and output Y:

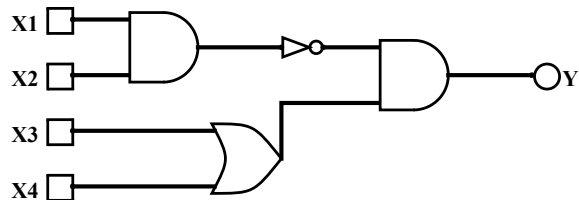


Calculate the output Y for the following values of the inputs:

X1	X2	X3	X4	Y
0	0	0	0	0 ✓
0	0	0	1	1 ✓
0	1	1	0	0 ✓
1	0	0	1	0 ✓
1	1	1	1	0 ✓

Simple Circuits 2

Given is this circuit with inputs X1, X2, X3, X4 and output Y:



For the following two questions, only enter a sequence of four 0s and/or 1s, without spaces or commas. E.g., X1=1, X2=1, X3=0, X4=1 must be entered as 1101.

Give an input sequence for X1, X2, X3, X4 that results in Y=0: ✓

Give an input sequence for X1, X2, X3, X4 that results in Y=1: ✓